## Experiment 7: Depth First Search (DFS)

### Aim

To implement the Depth First Search (DFS) algorithm for graph traversal.

### Objective

To traverse all the vertices of a graph using the Depth First Search technique.

### Algorithm

1. Start from the source vertex.
2. Mark the current vertex as visited.
3. Visit the current vertex.
4. Recursively visit all adjacent unvisited vertices.
5. Continue until all vertices are visited.
6. Display the traversal order.

### Flowchart

```
      Start
        |
        ▼
  Select Start Node
        |
        ▼
  Mark Node as Visited
        |
        ▼
  Visit Adjacent Node?
    ┌──────┴──────┐
    Yes      No
```

```
        |           |
        ▼           ▼
Visit Unvisited    End of Path
Neighbour            |
        |            ▼
        └──────────▶ Backtrack
                  |
                  ▼
                 End
```

**Python Code**

```python
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbour in graph[node]:
            dfs(neighbour)
```

```
dfs('A')
```

**Sample Output**

A B D E F C

**Result**

Thus, the Depth First Search (DFS) algorithm was successfully implemented and the graph was traversed using depth-first traversal.